UniDPP

DIGITAL PRODUCT PASSPORT · MULTI-BODY STEWARDSHIP

UNIDPP CONTRIBUTION

ISO/IEC JTC 5 · DIGITAL PRODUCT PASSPORT

DPP lifecycle, events and validity

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UniDPP — convened under CalConnect, multi-body stewardship

01 · SCOPE AND METHOD

From a mandated log to a model

EN 18223 §4.2 mandates change logging — each change recorded with identifier, timestamp, actor and changed properties — and §4.1.2.1 gives example values for `dppStatus` (active, inactive, archived, invalid) without an enumeration, a state machine or transition rules. EN 18246 §4.7 requires that every creation, modification and deletion be logged non-repudiablely with an identifiable editor. EN 18221 requires historical state be preserved and archived. What no document in the series supplies is the *model*: what kinds of things happen to a passport after first sale, who may append them, how status changes, and what was legally true at a given time.

This contribution supplies that model. It was derived by converting the full EN and CWA corpus and the IDTA submodel set to machine-readable form, extracting every lifecycle-relevant clause, and reconciling the three incompatible lifecycle taxonomies found in the series (the identifier granularity model of EN 18219, the nine-phase model of EN 18239 Annex A, and the ISO 14040-lineage model used by the EN 18223/18220/18219 terms). The proposal is offered as workstream B input and as the normative core of a prospective lifecycle standard.

Three design rules govern everything that follows: **nothing is ever edited in place** — every post-first-sale change is an appended event; **the event log is the authoritative record**, log-anchored and distributed; and **corrections never rewrite** — a correction is a new event that supersedes a prior value while preserving it in the log.

02 · THE EVENT TAXONOMY

Sixteen typed events

Each event type fixes what changes and which role appends it. Roles are those of EN 18239 (dealer, distributor, refurbisher, transfer of responsibility) extended with the framework's accredited-actor pattern for regulated claims.

ID	Event	What changes	Appender (role)
E1	custody.transfer	Custody link only (resale, lease, inheritance, liquidation).	Custodian + counterparty
E2	part.replace	Old child reference marked removed; new child reference added; bill-of-materials instance updated.	Repairer
E3	repair.perform	Repair record plus consumed parts (E2 within).	Repairer
E4	product.modify	Instance facts; may spawn a derived type (E4b) with mandatory profile re-evaluation.	Accredited modifier
E5	software.update / feature.unlock	Firmware/software version vector; capability enablement changes declared characteristics without physical change.	EO or service
E6	upgrade.install	New child reference.	Installer
E7	refurbish / remanufacture	Condition grade; remanufacture may be a transformation event producing a new type.	Refurbisher (EN 18239 role)
E8	consumable.replace	Child references, possibly regulated child passports (tyres, filters — recurring).	Any, per profile
E9	recall.campaign	Campaign reference; downstream traversal of the provenance DAG and custodian notification.	Regulator or EO
E10	correction (administrative)	New corrected fact with reason; prior value preserved in the log.	EO with attestation
E11	status.change	State-machine transition with authority check (suspend, invalidate, reinstate).	Regulator or EO per profile

E12	flag.security	Theft/loss blacklist entry; police report as a stamp.	Authority + registry
E13	decompose (end-of-life)	Inverse transformation, 1→N, producing material passports.	Recycler (accredited for claims)
E14	inspection.stamp	Lens-scoped attestation over subject state.	Any verifier
E15	milestone.record	Edge-segment state commitment (usage counters).	Device
E16	end-of-waste	Legal state transition by which waste ceases to be waste under applicable criteria — the passport-issuance boundary at which the material re-qualifies and a secondary-raw-material passport is issued.	Recycler; criteria per profile

Four cross-cutting rules bind the taxonomy. **Re-evaluation:** modifications and feature unlocks trigger re-running of the applicable profile's validation package, with a transitional "non-conformant pending re-evaluation" status and regulator notification per profile. **Recall propagation:** recalls traverse the provenance DAG downstream and notify custodians — no EN implements this; EN 18239 names take-back as a stage only — and the recall set is never enumerated centrally: the predicate is published, each custodian evaluates it locally (dark holdings under national procedure), and the initiator receives aggregates or nothing — predicate-based recall, not traversal-push enumeration. **Transformation:** splits, combinations and decompositions are typed events with quantity semantics — outputs carry input references with quantities and as-of state hashes, inputs transition to consumed/transformed while remaining verifiable for as-of queries, and mass balance is auditable. **Carrier replacement:** re-issuing a physical carrier (QR reissue) is a registry event on unchanged identity.

Installation is its own event semantics, not generic containment. An install event records child, parent, method, slot identity, pairing (none, firmware, key or module), any permanent alteration (soldered, welded, potted — possibly unrecoverable) and the performer: an installed, fastened component is a different legal object from the standalone one, with re-scoped warranty, parent-scoped slot identity and second-hand-once-removed status. Knowledge is bidirectional — the parent's manifest lists children; the child's log records its installation interval — because installation history is value-relevant provenance (a harvested battery sold standalone carries its in-vehicle history) and routes recalls, service access and end-of-life. A **recoverability spectrum** governs identity continuity: *restorable* (the child resumes standalone life), *harvestable* (recovered but altered — continues with harvested status and parent history), *destructive* (the child ceases; flows to a material passport as a transformation), *absorbing* (identity dissolves — recorded at finest granularity as dormant identifiers for later adoption). The rule: disassembly that restores a marketable object with identity continuity is installation; identity dissolution is a transformation event. Part replacement is accordingly **uninstall plus install** — two events, with the bill-of-materials instance updated.

03 · EN SUPPORT MATRIX

What the series provides today

The table below records, per EN of the series, the post-sale support found in the converted texts and the residual gap. It was compiled from full-text conversions and is reproducible from the published documents.

EN	Post-sale support in the text	Gap
18239	Best in series: lifecycle stages including maintaining, repairing, upgrading, refurbishing (Annex A); a repair-update example in the Introduction; roles with read-write rights (refurbisher §6.2); base roles from delegated acts (§5.2.8); delegation and emergency revocation.	Stages are a taxonomy, not operations: no event types, part swaps, firmware changes, derived types, theft handling or corrections.
18223	§4.2 acknowledges change of property and addition; logging fields (identifier, timestamp, actor, changed properties); example `dppStatus` values.	No event kinds, no transitions, no child references, no software/hardware versioning.
18246	§4.7: any creation, modification or deletion logged non-repudiablely, editor identifiable; compromised-data flagging.	Generic logging only; no event semantics.
18221	Historical state preservation and archiving of every change; persistence beyond the operator.	Change-kind agnostic (maps to the framework's Tier C).
18219	Identifier persistence across ownership change.	No identifier lifecycle events; battery/child swap unaddressed.
18220	—	Carrier replacement and reissue absent.
18216	—	Transport security only.
18222	Life Cycle API exists in outline (§4.1–4.8) plus Registry API.	Generic operations; no event vocabulary.
Context	EU right-to-repair Directive (EU) 2024/1799; IDTA content submodels 02018/02036/02048/02007/02019 cover *content* (maintenance instructions, change notifications, software nameplate).	Content is not passport mechanics: no EN connects repair events to passport state.

The verdict is factual, not rhetorical: the series *acknowledges* the post-sale world — EN 18239's stages, EN 18223 §4.2, EN 18246 §4.7 — and specifies **zero operations**. The taxonomy of §2 is proposed as the

normative content that closes this gap.

04 · THE STATE MACHINE

Legitimate status transitions

`dppStatus` becomes a closed enumeration with defined transitions, each carrying an authority check:

- **active** — the normative operating state; entered at issuance (issuer role) or reinstatement (E11, authority role).
- **suspended** (transitional "non-conformant pending re-evaluation" included) — entered by E4/E5 re-evaluation or regulator action; exit is active (re-evaluation pass) or invalid.
- **inactive** — the subject is not currently placed on the market (e.g., storage, export); reversible.
- **invalid** — terminal for the version: authority-checked E11 (regulator or EO per profile), misissuance findings, or adjudicated fraud; the event log remains verifiable.
- **archived** — Tier C archival state (EN 18221/OAIS); read-only, retained for the legally required horizon.

Two rules keep the machine honest. Status changes are **events** (E11), never field writes: the log records who held the authority and under which profile. And **profile growth never changes identity**: adding a new jurisdiction's profile to an existing product is a registry applicability event with an effective window — including retroactive flags where the law says so — not a new passport.

05 · TIME AND VALIDITY

ISO 8601, ISO 34000, as-of queries

The time basis is the one the series already cites normatively: **ISO 8601-1 and ISO 8601-2, including Amendment 1** (time deviations), with **ISO 34000** supplying the event vocabulary. All timestamps are UTC with explicit offset handling; the framework's test suite rejects the defects observed in the corpus (an Annex B example in EN 18223 prints an invalid timestamp using en-dashes in the very clause that teaches the format — validator-enforced, not publication-hoped).

Validity is modelled with two authoritative query modes, because the law needs both:

- **Live view** — the current state of the passport, with every element labelled by freshness (staleness is a displayed verdict, never silence).
- **As-of snapshot** — the notarized, log-anchored reconstruction of what was legally true at time T: "was this product compliant with profile P when sold in 2028?" As-of answers are reproducible because event order is total in the anchored log and registry versions are pinned per validation.

Between the two sit the **three verification readings** defined in paper 5 (evidentiary, current-state, cryptographic); verdicts state which reading they answer. Retroactivity of revocation follows the reason taxonomy there: prospective reasons preserve pre-event as-of validity; retroactive reasons (misissuance, fraud) void it ab initio within an explicit window.

Freshness requirements are gated by capability class (paper 1, §04): profiles declare, per data element, the required truth mode (attest-sampled, self-committed, or both with precedence), and the requirement must be satisfiable by the class — an S0 (silent, testimony-only) product cannot satisfy a live-freshness profile.

06 · REFERENCES

1. EN 18223 §4.1.2.1, §4.2; EN 18239 Annex A, §5.2.8, §6.2; EN 18246 §4.7; EN 18221; EN 18219; EN 18222 §4.1–4.8.
2. ISO 8601-1:2019 and ISO 8601-2:2019 including Amd 1; ISO 34000 (event vocabulary).
3. Directive (EU) 2024/1799 (right to repair); CWA 18291 (actor–event catalogue).
4. ISO/IEC 15459 (identifiers); OAIS (ISO 14721) reference model for archival.
5. Companion papers 1, 2 and 5 of this series.

Drafting: CalConnect — in the UniDPP framework (multi-body stewardship; Apache-2.0/MIT per repository).

Paper 3 of 6 · dpp-lifecycle-events-validity.html · 2026-09-07